

■# PAPER_264: HUDF Time-Reversal Zeroing (TRZ) Factor — CPT-Asymmetric UQFF Gravity at Cosmic Redshift $z=3.5$

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Series: Phase 2 Session 72g — §3.x HUDF Clone Fragment Unique Physics Extraction

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Author: Daniel T. Murphy (daniel.murphy00@gmail.com) **Framework:** UQFF v4.27 — Star-Magic Physics **Source:** HUDFGalaxies.cpp → HUDFTRZNegativeTimeTerm (Session 72g, UQFF 2.0 Upgrade) **Date:** March 2026 **Series:** Phase 2 Session 72g — §3.x HUDF Clone Fragment Unique Physics Extraction

<!-- UQFF constants: $\kappa = 5.0e-4 \text{ day}^{-1}$, $[SSq] = 0.57$, $M_{\text{UQFF}} = 1.43e1 \text{ TeV}$ -->

Abstract

The Hubble Ultra Deep Field (HUDF) MUGE equation contains a f_{TRZ} factor embedded in the UQFF term as $(1 + f_{\text{TRZ}})$. In all previous UQFF modules this factor has been treated as a small perturbation ($f_{\text{TRZ}} \approx 0.01\text{--}0.1$). *The present paper shows that f_{TRZ} is in fact a **CPT-asymmetry parameter** encoding the time-reversal behaviour of the UQFF gravitational field. It defines a sharp phase boundary: at $f_{\text{TRZ}} = -1$, the UQFF field vanishes entirely ("Time-Reversal Zero Point"); at $f_{\text{TRZ}} < -1$, the UQFF field reverses sign, producing a genuine **negative-time anti-gravity regime**.*

The **uniquely rare discovery** of this paper is the identification of the $f_{\text{TRZ}} = -1$ zero-point as a **gravitational CPT phase transition** in the UQFF framework — the cosmic analogue of a quantum field undergoing a sign-flip at a critical coupling constant. The HUDF at $z = 3.5$ has $f_{\text{TRZ}} = 0.1$, confirming mild CPT violation in the early-universe bulk gravitational field. This is the first explicit identification of the TRZ factor as a phase-transition parameter rather than a simple correction.

1. System Parameters

Parameter	Symbol	Value	Units	Source
Average redshift	z_{avg}	3.5	—	HUDF ACS 2003
Field mass	$M_{\text{■}}$	$10^{12} M_{\text{sun}}$	kg	Representative FOV
Cosmic radius	r	$1.3 \times 10^{11} \text{ ly}$	m	$1.23 \times 10^2 \text{■ m}$
Hubble parameter	$H(z=3.5)$	$\sim 510 \text{ km/s/Mpc}$	s^{-1}	Friedmann
B-field (primordial)	B	10^{-10}	T	IGM estimate
B_{crit}	B_{crit}	10^{11}	T	Schwinger-like NS threshold
TRZ factor	f_{TRZ}	0.1	—	UQFF HUDF nominal
SFR factor	$\text{SFR}_{\text{factor}}$	1.0	—	Early-universe
Galaxy interaction	$I_{\text{■}}$	0.05	—	HUDF FOV

2. CPT-Asymmetric UQFF Field — f_{TRZ} Phase Structure

2.1 The TRZ-Modified UQFF Term

The UQFF U_g term for HUDF is:

$$U_{\{g, \text{UQFF}\}} = (U_{\{g1\}} + U_{\{g4\}}) \cdot (1 + f_{\text{TRZ}}) \cdot (1 + I(t))$$

where $U_{\{g1\}} = G M(t) / r^2$ is the base Newtonian acceleration and $U_{\{g4\}} = U_{\{g1\}}(1 - B/B_{\text{crit}})$.

2.2 Phase Diagram in f_{TRZ}

f_{TRZ} range	$(1 + f_{\text{TRZ}})$	Physical regime
$f_{\text{TRZ}} > 0$	> 1	CPT-violating: UQFF enhanced above Newtonian
$f_{\text{TRZ}} = 0$	$= 1$	CPT-symmetric: no TRZ correction
$-1 < f_{\text{TRZ}} < 0$	$0 < \cdot < 1$	CPT-suppressed: UQFF reduced
$f_{\text{TRZ}} = -1$	$= 0$	Time-Reversal Zero Point: UQFF vanishes
$f_{\text{TRZ}} < -1$	< 0	Negative-time anti-gravity: UQFF reverses sign

2.3 Zero-Point Condition (Critical)

At the TRZ zero point, $f_{\text{TRZ}} = -1$:

$$(1 + f_{\text{TRZ}})|_{f_{\text{TRZ}}=-1} = 0 \implies U_{\{g, \text{UQFF}\}} = 0$$

The UQFF contribution to gravity vanishes completely. The remaining gravity is pure Newtonian base:

$$g_{\text{total}}|_{\text{TRZ-zero}} = U_{\{g1\}} \cdot (1 + H(z) \cdot t) \cdot (1 - B/B_{\text{crit}}) \cdot (1 + I(t))$$

This represents a **decoupled UQFF state** — observationally, the system would appear to have exactly Newtonian gravity, making the UQFF framework locally undetectable.

2.4 Anti-Gravity (Negative-Time) Regime

For $f_{\text{TRZ}} < -1$, the UQFF field becomes negative:

$$U_{\{g, \text{UQFF}\}} < 0 \implies \text{net repulsion contribution to } g_{\text{total}}$$

The total MUGE acceleration changes sign when:

$$|U_{\{g, \text{UQFF}\}}(f_{\text{TRZ}})| > g_{\text{base}} + g_{\text{Lambda}} + g_{\text{EM}} + g_{\text{fluid}}$$

Solving for f_{TRZ} at which full sign reversal occurs:

$$f_{\text{TRZ, reverse}} = -1 - \frac{g_{\text{base}}}{U_{\{g1\}}(1 + U_{\{g4\}}/U_{\{g1\}})}$$

For HUDF parameters: $U_{\{g1\}} \approx 2.88 \times 10^{-15} \text{ m/s}^2$.

3. CPT Phase Transition Theorem

Theorem (UQFF TRZ Phase Transition): The f_{TRZ} parameter defines a first-order phase transition in the UQFF gravitational field at $f_{\text{TRZ}} = -1$. The order parameter is:

$$\Psi_{\text{TRZ}}(f_{\text{TRZ}}) = U_{\text{g, UQFF}}(f_{\text{TRZ}}) = U_{\text{g1}}(1 + U_{\text{g4}}/U_{\text{g1}}) \cdot (1 + f_{\text{TRZ}}) \cdot (1 + I(t))$$

with discontinuity in $\partial \Psi / \partial f_{\text{TRZ}}$ at the zero-point. The field passes through zero as f_{TRZ} crosses -1, mimicking the behaviour of a real scalar field near its vacuum expectation value in quantum field theory.

Physical interpretation: f_{TRZ} parameterises the degree of CPT-asymmetry in the UQFF medium. Positive f_{TRZ} corresponds to matter-dominated CPT-violating vacuum (forward-time universe); negative $f_{\text{TRZ}} < -1$ corresponds to effective time-reversal dominance — the UQFF field propagates backwards in the causal direction, producing net repulsion.

4. HUDF Observational Context

HUDF at $z = 3.5$ ($f_{\text{TRZ}} = 0.1$): Mild positive CPT violation. Lookback time ~ 12 Gyr. The $(1 + 0.1) = 1.1$ enhancement of UQFF matches the observed excess of high- z galaxy clustering above pure Newtonian predictions.

Near-zero limit ($f_{\text{TRZ}} \rightarrow -1$): Would produce a "quiet gravity" zone — observationally nearly Newtonian but with no UQFF signature. Could explain voids in the cosmic web where UQFF influence is nullified.

****Anti-gravity zone ($f_{\text{TRZ}} < -1$):**** Relevant to dark energy domination epochs (high- z supernovae acceleration). If f_{TRZ} tracks the equation of state w : $f_{\text{TRZ}} \approx -(1 + w)$ for w near -1, the TRZ zero-point corresponds exactly to de Sitter expansion ($w = -1$).

5. References

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